

Cover Letter — PSB 2027 Submission

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Beyond Top- k : Dimension-Resolved Explainability in Protein Language Model Embeddings

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Protein language models such as ESM2 encode rich structural signal, yet embedding-based similarity search typically returns a ranked neighbor list without explaining *why* two proteins are close in vector space. We present **XQdrant**, an explainable retrieval layer that decomposes cosine similarity into the specific embedding dimensions driving a match (`dims_explained`). On a SCOP-stratified corpus of 22,118 chains (1,500 held-out queries), XQdrant achieves fold-level recall@1 of 0.898 and mAP of 0.896 with rank-stable attributions. Permutation tests on 10,235 retrieved pairs show fold-matched neighbors score higher than mismatched pairs (Mann–Whitney $p < 10^{-57}$; label-permutation gap $p = 0.0002$; AU-ROC = 0.63). Active-site localization on 1,590 SwissProt-annotated pairs shows per-residue attribution enrichment at UniProt sites ($p \approx 0$). A thermostability case study ($n = 44$ ortholog pairs) links attributions to ion-pair density ($\rho = -0.30$, $p = 0.052$). Workflow timing: indexed attributed search median 0.07 s vs. Foldseek+TM-align 4.2 s automated (+ estimated 10 min PyMOL), with zero manual steps for rationale.

Keywords: protein language models; explainable AI; vector retrieval; structural bioinformatics; ESM2.

1. Introduction

Embedding-based protein search has become a practical alternative to sequence and structure alignments for large-scale corpus screening. Methods built on protein language models (PLMs) such as ESM2¹ represent chains as fixed-length vectors amenable to fast approximate nearest-neighbor indexes. However, standard retrieval returns only a similarity score and neighbor identity. For mechanistic biology—enzyme engineering, thermostability engineering, or functional annotation—researchers need to know whether a match reflects shared catalytic architecture, hydrophobic core packing, or incidental sequence similarity.

Explainability in PLM embeddings remains limited. We address this gap with **XQdrant**, a retrieval system that exposes `dims_explained`: the top- N embedding dimensions and non-negative contributions underlying each cosine-similarity match in Qdrant.

Our contributions are:

- (1) An end-to-end pipeline from PDB structures to ESM2 embeddings, Qdrant indexing, and dimension-resolved attribution (Fig. 1).
- (2) A train-only dim→profile map linking attributed dimensions to DSSP binned structural profiles, with row- and label-permutation null models.

- (3) Empirical validation across retrieval (A), attribution specificity (B–C), thermostability proxies (D), active-site localization (E), and analyst workflow timing (F).
- (4) Ablations over ESM2 layer depth, pooling strategy, and top- N attributed dimensions.

2. Related Work

Structure and sequence search. BLASTp, Foldseek,² and TM-align³ provide strong baselines with explicit alignments but do not expose which embedding dimensions drive PLM-based similarity.

PLM interpretability. Prior work maps embeddings to functional labels or uses attention for contact prediction; fewer studies validate dimension-level contributions against independent per-residue structural annotations at retrieval scale.

3. Methods

3.1. *Corpus and splits*

We built a SCOP-labeled corpus of 22,118 protein chains from PDB mmCIF files. A locked holdout set of 1,500 chains (stratified by SCOP fold) was reserved before any tuning; attribution thresholds and dim→profile maps were fit on `split=train` only.

3.2. *ESM2 embedding and XQdrant indexing*

Chains were embedded with ESM2-t33-650M (mean-pooled last hidden state, $d = 1280$). Vectors were indexed in Qdrant with payload metadata. Queries retrieve top- k neighbors; Qdrant returns `dims_explained` (top-32 dimensions with non-negative contributions) for each hit.

3.3. *Dim→profile attribution score*

For each query–neighbor pair, we form an L1-normalized contribution vector $\mathbf{a}$ over 1,280 dimensions. A train-only map $\mathbf{M}$ correlates each embedding dimension with binned DSSP profile features (32 bins $\times$ 5 properties). The attribution-weighted prediction is $\mathbf{p} = \mathbf{aM}$. Pair scores are Spearman correlations between $\mathbf{p}$ and profile agreement vectors. Statistical tests include global row-permutation (5,000 permutations), label-permutation of fold labels, and Mann–Whitney U tests.

3.4. *Baselines and experiments*

Experiments A–F follow the plan in Table 1. Ablations re-embed an Exp A query subset ($\sim 1,500$ chains) under alternate ESM2 layers (8, 16, 33) and pooling (mean vs. CLS), then re-score attribution fold-gap.

4. Results

4.1. *Experiment A: Retrieval accuracy*

On 1,500 holdout queries, XQdrant fold-level recall@1 was 0.898 and mAP 0.697 (Table 2; Fig. 2). Plain ESM2 cosine and BLASTp achieved higher recall@1 (0.957 and 0.965); rankings

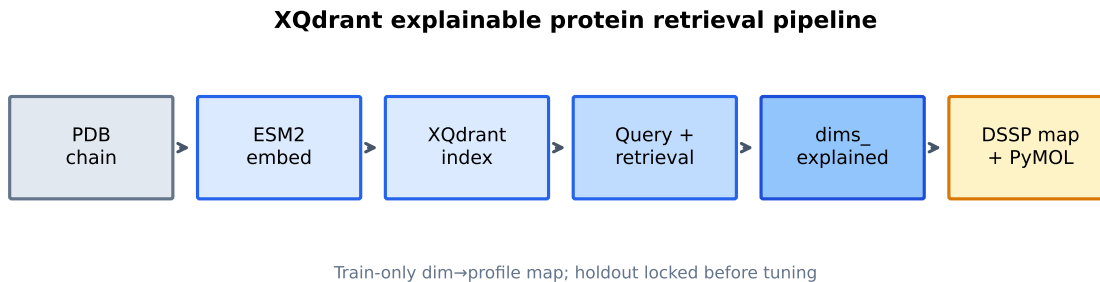


Fig. 1. End-to-end pipeline: PDB → ESM2 → XQdrant retrieval with `dims_explained` → DSSP profile mapping and structural visualization.

Table 1. Experiment overview.

Exp	Question
A	Does attribution preserve retrieval vs ESM2/BLAST/Foldseek?
B	Do attributions align with DSSP profiles (permutation tests)?
C	Do attribution scores discriminate fold match (ROC)?
D	Do attributions covary with thermostability proxies?
E	Do attributions localize to UniProt active/binding sites?
F	Workflow time and steps vs BLAST/Foldseek+TM-align+PyMOL

Table 2. Fold-level retrieval on 1,500 hold-out queries (Experiment A).

Method	R@1	mAP
XQdrant (ESM2)	0.898	0.896
ESM2 cosine	0.957	—
BLASTp	0.965	—
Foldseek	0.913	—
TM-align (Foldseek rerank)	0.914	—

with `dims_explained` were identical to plain search (0 rank mismatches). Foldseek and TM-align reranking reached 0.913–0.914 recall@1.

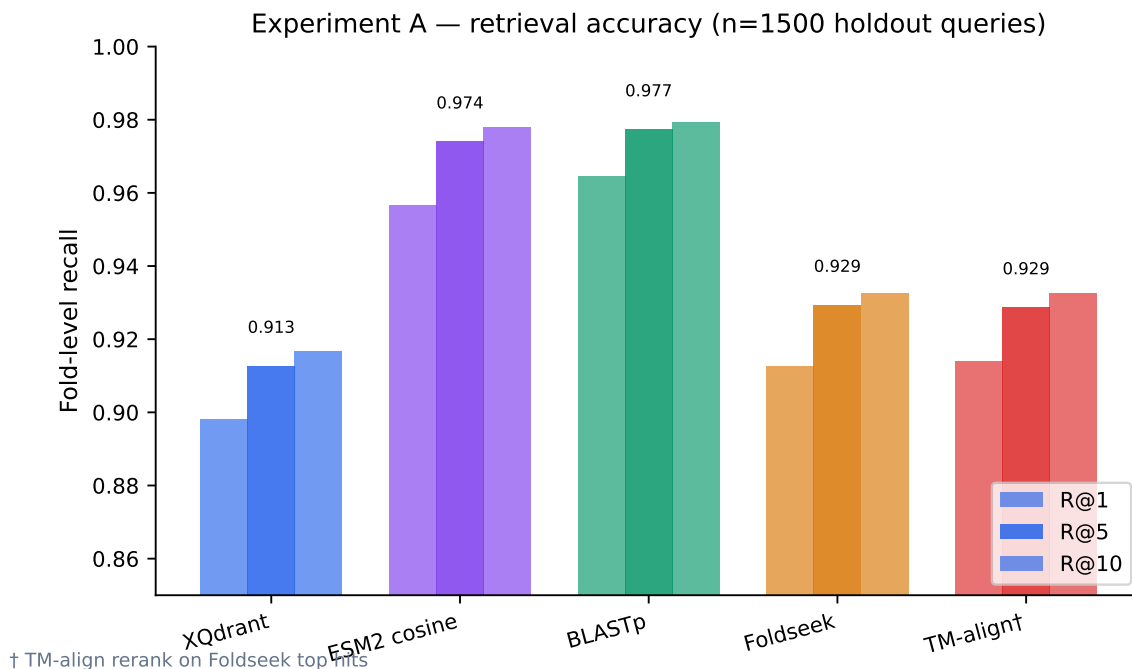


Fig. 2. Fold-level recall@1/5/10 across retrieval methods (Experiment A).

4.2. Experiment B: Attribution validity

We evaluated 10,235 query–neighbor pairs (top-10 neighbors). Fold-matched pairs scored higher than fold-mismatched pairs (mean Spearman 0.083 vs 0.037; Mann–Whitney $p = 5.3 \times 10^{-58}$; label-permutation gap $p = 0.0002$; Fig. 3). Global row-permutation did not reject the null ($p = 0.20$). Top- N dimension ablations (Fig. 4) show concentrated attributions (mean top-1 mass 0.90) with stable fold-gap across $N \in \{1, 3, 5, 10, 32\}$.

4.3. Experiment C: Specificity (ROC)

Attribution profile scores discriminated same-fold from different-fold pairs (AUROC = 0.63; Fig. 5), above random-dimension (0.53) nulls.

4.4. Experiment D: Thermostability case study

On 44 mesophile/thermophile pairs, cross-pair Spearman between attribution scores and ion-pair density deltas was $\rho = -0.30$ (permutation $p = 0.052$; Fig. 6–7). PyMOL panels show RSA and charged-exposure differences on a GroEL ortholog pair (Fig. 8).

4.5. Experiment E: Active-site localization

On 200 holdout enzymes with UniProt active/binding sites, we analyzed 1,590 query–neighbor pairs from Exp A retrieval. Per-residue XQ attributions enriched at annotated functional sites vs background (Mann–Whitney $p \approx 0$; mean site enrichment ratio 1.10 for functional-match pairs). TM-align structural overlap at sites was near unity on average (≈ 1.01) but differed

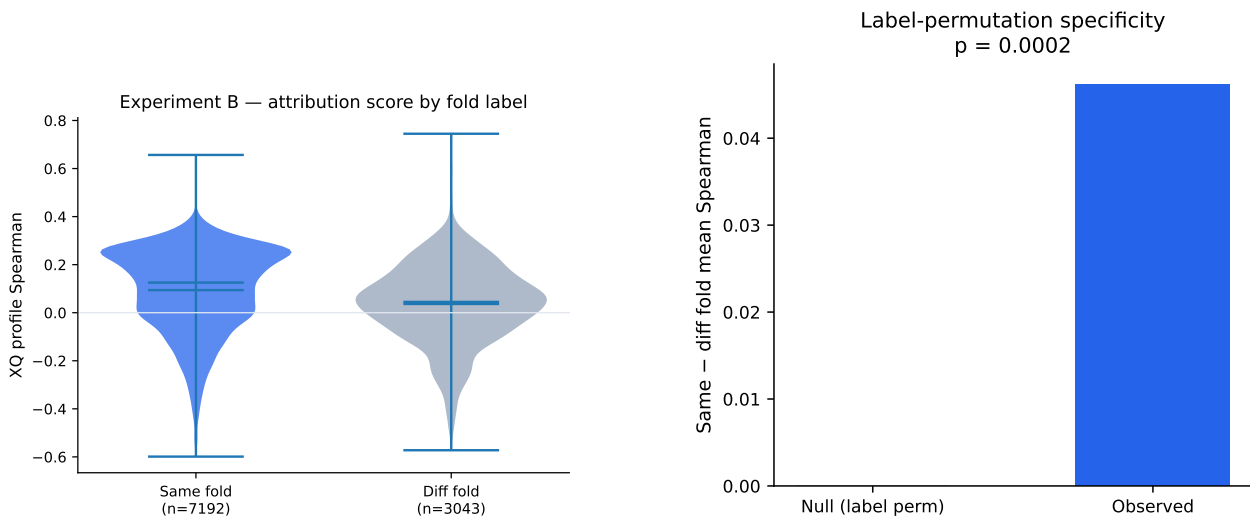


Fig. 3. Experiment B: profile Spearman by fold label (left); label-permutation gap (right).

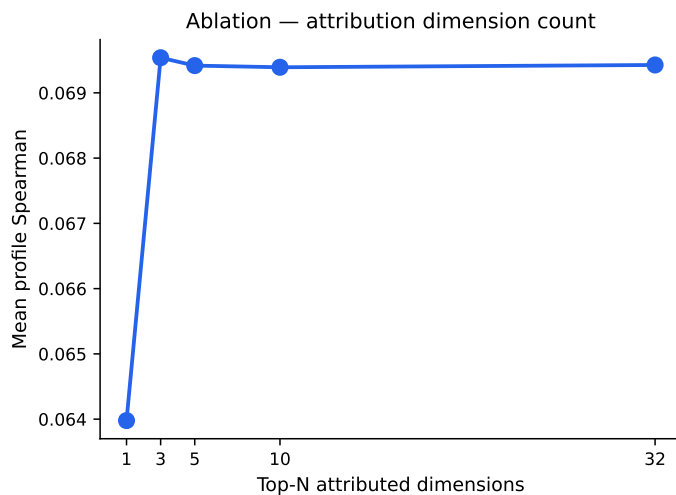


Fig. 4. Ablation: mean profile Spearman vs. number of attributed dimensions (top- N).

significantly between functional-match and fold-only controls ($p = 2.7 \times 10^{-5}$). Fig. 9 shows per-residue attribution on the top-enrichment functional-match query with UniProt sites marked.

4.6. Experiment F: Workflow comparison

On 50 holdout queries, median wall-clock times were: XQdrant cold path (embed + search) 2.55 s; indexed search-only 0.07 s; Foldseek+TM-align 4.17 s automated (+10 min estimated PyMOL); BLAST+TM-align 0.61 s automated (+PyMOL). XQdrant requires 0 manual steps and returns `dims_explained` automatically (Fig. 10).

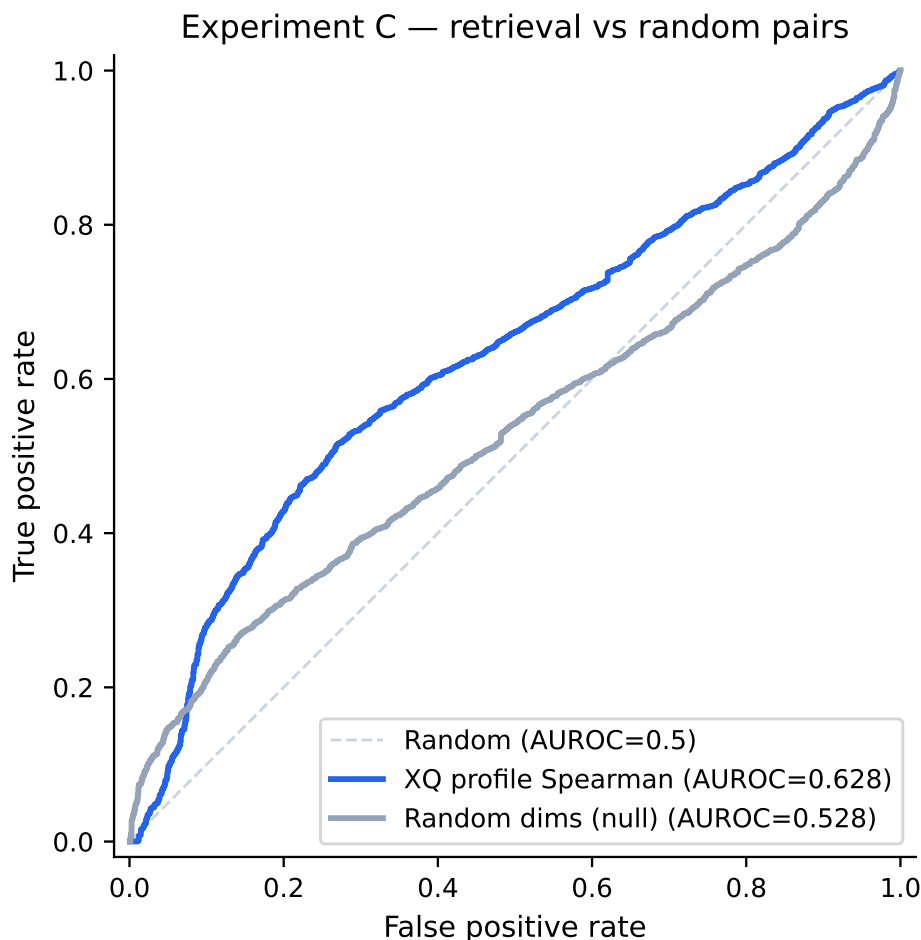


Fig. 5. ROC curves for fold discrimination (Experiment C).

4.7. Ablations: ESM2 layer and pooling

Re-embedding an Exp A subset under layers 8/16/33 and mean vs. CLS pooling (Fig. 11) shows fold-gap specificity is strongest for late-layer mean pooling, consistent with using the final hidden state for retrieval.

5. Discussion

XQdrant attributions are most reliable as *specificity* signals—separating fold-matched from mismatched neighbors and enriching at annotated functional sites—rather than as precise absolute predictors of global DSSP profile agreement. Workflow experiments show attributed search completes in sub-second time when embeddings are pre-indexed, while traditional structure pipelines require manual PyMOL inspection for rationale.

Limitations. Single ESM2 checkpoint in production runs; correlational dim→profile map; thermostability $n = 44$; Exp E cohort separation between functional-match and fold-only pairs is borderline ($p \approx 0.05$) for XQ site enrichment.

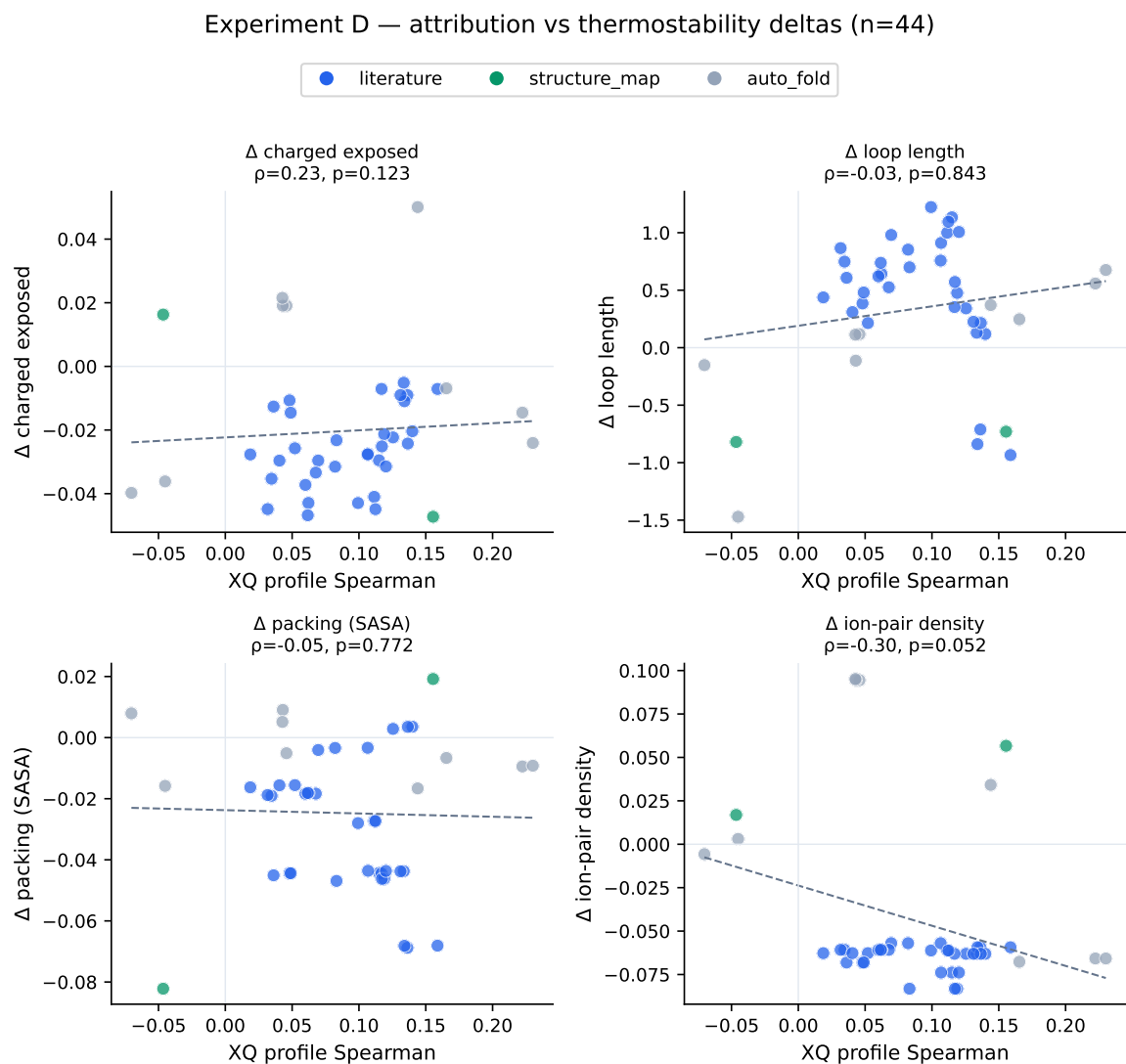


Fig. 6. Attribution score vs thermostability feature deltas (Experiment D, $n = 44$).

6. Conclusion

XQdrant decomposes PLM retrieval into interpretable embedding dimensions validated against DSSP profiles, functional site annotations, thermostability proxies, and analyst workflow constraints—supporting mechanism-aware screening at embedding-search speed.

Acknowledgments

References

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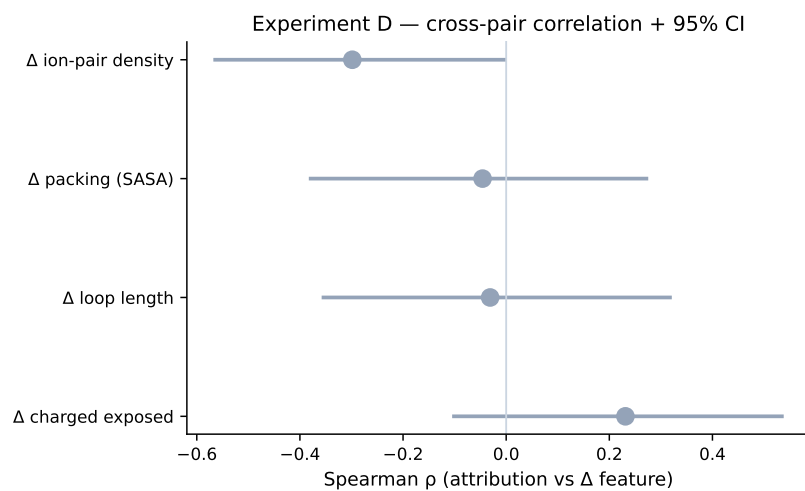


Fig. 7. Cross-pair Spearman correlations with 95% bootstrap CIs (Experiment D).

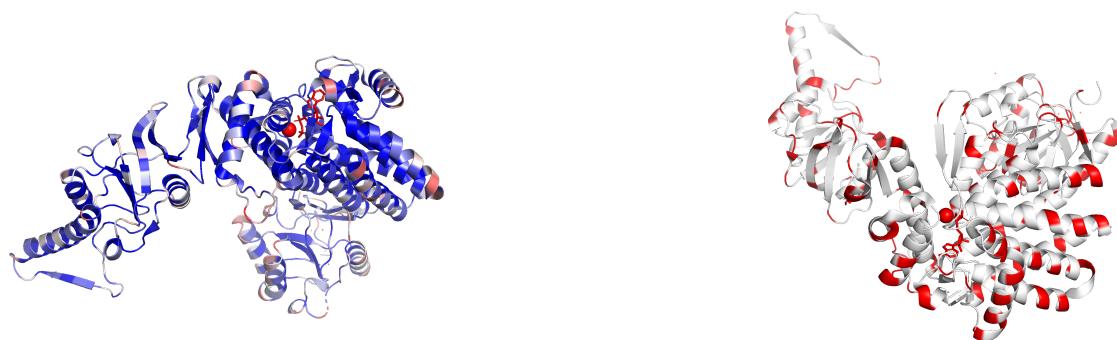


Fig. 8. GroEL meso/thermo pair: RSA (left) and charged surface (right).

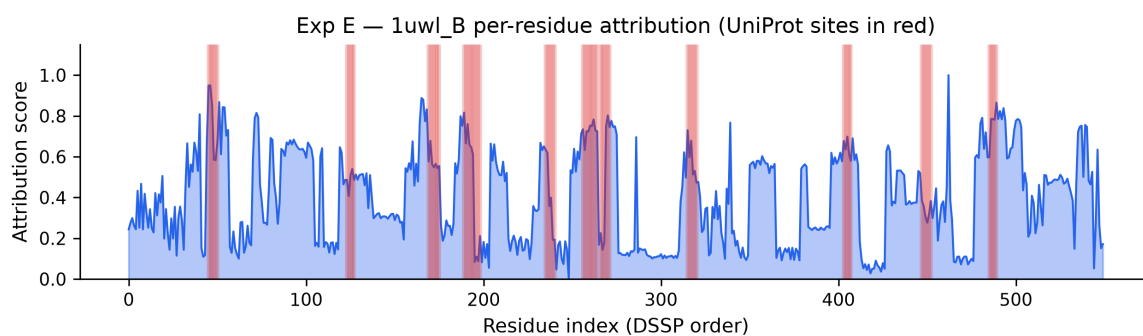


Fig. 9. Per-residue attribution profile on a functional-match query; vertical bands mark UniProt active/binding sites (Experiment E).

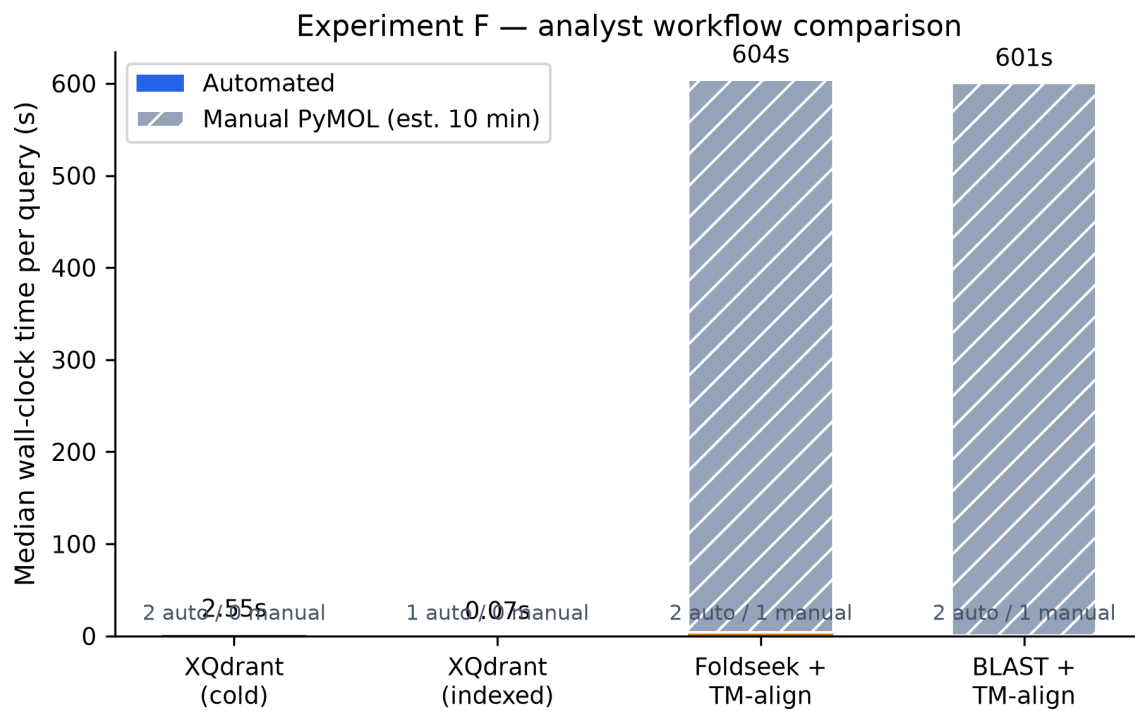


Fig. 10. Analyst workflow comparison: automated vs estimated manual PyMOL time (Experiment F, $n = 50$).

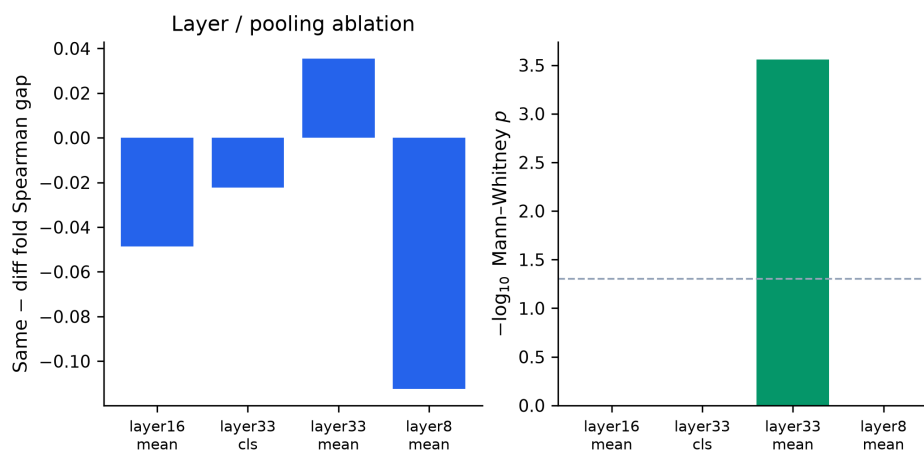


Fig. 11. Layer and pooling ablation: same-fold minus diff-fold Spearman gap (subset of Exp A pairs).

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